

Article

A spherical microphone prototype for multichannel recording: Technological design, artistic applications and compositional implications

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Abstract

Recent developments in spatial audio and immersive technologies have expanded creative possibilities for composers and sound artists. This article presents a novel prototype of a spherical microphone with an ellipsoid casing and ten motorised condenser capsules, each capable of real-time adjustment of orientation and polar pattern. Unlike fixed-pattern or conventional ambisonic arrays, this design enables dynamic control over spatial coverage and directivity, offering new opportunities for multichannel recording, live performance and interactive sound art. While software-based spatialisation offers some flexibility, physical reconfiguration of capsules provides superior responsiveness and avoids latency, phase artefacts or resolution loss. This is especially critical in performance contexts where immediate acoustic adaptation is required. The system allows direct manipulation of capsule parameters during rehearsal or installation, effectively transforming the microphone into a performative instrument. The article compares the prototype with existing commercial ambisonic microphones, highlighting its distinctive advantages in workflow and compositional strategy. Use-case scenarios demonstrate how real-time control over spatial parameters enhances both technical precision and artistic expressiveness. The article concludes with a discussion of future directions, including collaborative testing with practitioners and integration into creative environments where spatial transparency, fidelity and interactivity are essential.

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1. Introduction

Spatialisation has long been a central concern for electroacoustic composers and sound artists, shaping the interplay between technology, performance and perception (Robusté 2018: 296–307; Landy 2007; Smalley 2007: 35–58; Bates 2009). As immersive media – including virtual reality (VR) applications, 3D cinema and site-specific installations – continue to proliferate, there is an increasing demand for recording set-ups that accurately and flexibly capture multidimensional sound fields (Rafaely 2005, 135–43; 2008a: 5–8). This evolution enables composers and listeners alike to conceptualise space not merely as a presentation medium but as a critical compositional element, intricately woven into musical form, texture and narrative (Wishart 1996; Landy 2007; Young 2007: 25–33).

Recent advances in VR audio processing techniques have further broadened the possibilities for immersive and realistic sonic experiences in video games, training simulations and virtual concerts. Although early developments in audio recording lagged behind breakthroughs in image processing, the past few years have witnessed a surge in innovative devices designed to enhance spatial perception. Consequently, current research in music technology and sound studies increasingly underscores the

synergy between emerging solutions – such as ambisonics and binaural rendering – and aesthetic innovation, emphasising listener agency, interactive performance and the fluid boundaries between recorded and synthesised sonic environments (Rebelo 2003: 181–6; Bates 2009).

In response to these technological and artistic developments, this study introduces a prototype spherical microphone system that integrates second-order ambisonics, binaural capture and real-time parametric control. This integrated approach bridges engineering design with artistic practice in electroacoustic and interactive music composition. Furthermore, ongoing research highlights the compositional significance of enhanced spatial precision, fostering an enduring dialogue between acoustic measurement and creative practice (Smalley 1997: 107–26; Bates 2009; Ramakrishnan 2009: 268–76).

2. Review of literature and applied devices in sound recording

Spherical microphone arrays have become fundamental components in advanced sound recording systems, as well as in spatial data collection, sound wave modelling and 3D audio imaging. Designing these arrays requires careful consideration of multiple acoustic principles, design criteria and performance guidelines to ensure optimal accuracy and functionality (Rafaely 2005: 135–43; 2008a: 5–8; Pollow, Behler and Masiero 2009; Hagaia et al. 2011: 2003–15; Lee and Johnson 2021: 871–87). Such arrays enable the

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capture and analysis of complex sound fields, which is crucial for reconstructing three-dimensional auditory environments (Yan et al. 2011: 361–71).

A key aspect of constructing these devices is the application of mathematical models based on the head-related transfer function (HRTF). HRTF describes how sound waves are modified by a listener's anatomical features – such as the auricles, ear canals, torso, arms and head – before reaching the eardrums (Gardner and Martin 1994; Kahana et al. 1999: 1503–16; Huang and Benesty 2004; Hardera et al. 2006: 39–46; Vidal et al. 2021: 1–8). By replicating essential psychoacoustic cues, including interaural time difference, interaural level difference and pinna filtering, these devices allow composers to create intricate ‘sonic illusions’ in which sounds are perceived as emanating from precisely defined spatial locations (Moore 2016; Findlay-Walsh 2017: 121–30).

However, conventional dummy-head microphones rarely support multichannel or higher-order ambisonic (HOA) capture, limiting their application in hybrid performance contexts – such as scenarios where loudspeaker arrays and headphone-based binaural sound coexist for the same audience (Blesser and Salter 2006; Bates 2009). Academic discourse on binaural listening emphasises its capacity to foster intimate and personalised encounters with sonic art (Bates 2009; Findlay-Walsh 2017: 121–30). Consequently, many composers have sought a single, versatile device capable of capturing both the anthropomorphic spatiality of binaural recordings and the multidirectional, immersive perspective afforded by ambisonic techniques (North 2024: 292–302). This gap in current technological solutions is the primary motivation for the development of our spherical microphone prototype.

One of the most dynamic areas of contemporary music and sound art is the diversity of presentation settings and the increasingly complex experiences available to listeners engaging with spatial audio. From traditional concert venues with multichannel loudspeaker arrays, through headphone-based binaural installations, to hybrid environments in virtual and augmented reality (AR), each setting poses distinct challenges and possibilities for both composers and audiences.

Presentation workflows have evolved alongside these settings. In a multichannel concert scenario, the placement of speakers and the spatial diffusion of sound are carefully choreographed to create immersive sound fields, allowing audiences to experience movement, localisation and envelopment in real space (Bates 2009; Baalman 2010: 209–18). By contrast, binaural installations – whether in galleries, museums or site-specific environments – invite listeners to inhabit intimate, individualised auditory spaces. Here, headphone listening circumvents the need for complex loudspeaker arrays but requires precise control of HRTF and a heightened sensitivity to perceptual cues (Findlay-Walsh 2017: 121–30; Blesser and Salter 2006).

Recent technological advances have further diversified the workflows available to both practitioners and listeners. Virtual reality concerts, for example, immerse participants in entirely simulated environments, where spatial audio engines dynamically adapt to head movements and interactive actions, producing experiences that blend the boundaries between real and virtual (Jerald 2015; Grimshaw 2014). Similarly, AR and mixed-reality (MR) artworks enable mobile, location-aware soundscapes that respond to audience position or behaviour in real time, fostering active engagement and participatory modes of listening (Bull and Back 2015).

These developments have prompted new compositional strategies. Composers and sound designers must now consider not only how to sculpt sound in space but also how to anticipate

and accommodate the embodied experience of diverse listeners – whether seated in a concert hall, moving through an installation or navigating a virtual world. This requires workflows that integrate spatial recording, real-time sound manipulation, interactive control and detailed evaluation of audience response. Iterative feedback from test audiences and participatory workshops has become an important component of artistic development, ensuring that spatial works remain perceptually compelling across multiple presentation formats.

By enabling flexible, real-time configuration of spatial recording parameters, the spherical microphone prototype directly addresses these evolving workflows. Its capacity for simultaneous ambisonic and binaural capture, dynamic reorientation and adaptive directivity supports creative experimentation with a wide range of presentation settings, bridging the gap between fixed studio production and responsive, interactive environments. As such, the device is designed to facilitate not only technical excellence in spatial audio capture but also the rich diversity of listener experiences that define contemporary electroacoustic art.

The choice of playback format has a profound impact on listener experience and the creative strategies available to composers and sound artists. Binaural playback, typically delivered via headphones, offers highly personalised spatial experiences by simulating the human ear's reception of sound, including HRTF. This method is especially effective in installation contexts, VR applications and scenarios where individualised perception and portability are paramount. By contrast, ambisonic recordings are decoded for playback over loudspeaker arrays in concert halls, galleries or immersive environments, enabling shared spatial experiences and dynamic sound field reconstruction within a given space. While HOA can be rendered binaurally for headphones, and binaural signals can be artificially spatialised for loudspeakers, each approach has distinct perceptual and technical implications. For instance, ambisonic playback is well-suited to large-scale, multilistener events, supporting localisation and envelopment for an entire audience, whereas binaural techniques excel in situations demanding intimate, highly resolved spatial detail for a single listener. The spherical microphone prototype's dual-output capability facilitates both workflows, allowing for seamless transitions between individual and collective listening modes in electroacoustic composition, installation art and virtual environments (Zaunschirm, Frank and Zotter 2020; Blesser and Salter 2006).

While a number of commercial ambisonic microphones are available – such as the Eigenmike EM32, Zylia ZM-1, Soundfield SPS200 and Core Sound TetraMic – each of these devices is limited in terms of either fixed capsule geometry, lack of real-time mechanical adjustability or restricted directivity options. The Eigenmike EM32, for example, provides a dense spherical array with fixed omnidirectional capsules, enabling HOA recording but offering no physical reconfiguration during use. The Zylia ZM-1, though compact and user-friendly, is similarly limited to a static capsule array and primarily supports first-order ambisonics. Soundfield and Core Sound devices typically offer four fixed capsules, with digital processing providing virtual rotation and decoding.

In contrast, the spherical microphone prototype described here introduces two key innovations: (1) each capsule is motorised and can be physically reoriented in real time, and (2) polar patterns can be selected independently per capsule, enabling the user to adapt both spatial coverage and directional focus to the recording scenario at hand. This level of dynamic configurability is not available in existing commercial products and opens up new

artistic possibilities for spatial audio recording, live performance and interactive sound art. The device is therefore intended not as a direct competitor to laboratory-standard HOA microphones but as a flexible and creative tool tailored to the needs of composers, sound artists and practitioners working at the intersection of technology and musical expression.

Devices developed for binaural, spatial and immersive audio capture typically fall into one of the following categories:

Human Head with Torso:

Life-sized human head models mounted on a partial or complete torso, equipped with anatomically accurate auricles. These models are often produced from plastic composites or via 3D printing using materials with varied acoustic properties to achieve high-precision recordings (Gardner and Martin 1994; Mobashsher and Abbosh 2014: 1401–4; Uchibori et al. 2015: 43–5; O'Connor and Kennedy 2021; Ting et al. 2021; Vidal et al. 2021: 1–8).

Human Head without Torso:

Standalone head models featuring realistic auricles to simulate natural spatial perception, albeit without the acoustic effects introduced by a torso (Toshima, Uematsuy and Hirahara 2003: 327–9; Hardera et al. 2006: 39–46; Obadijah 2021: 87–94; Ting et al. 2021; Grandjean et al. 2023).

Head Models without Auricles:

Simplified head designs that omit external ear structures, primarily used in studies investigating dynamic binaural cues and the impact of head movements on spatial sound localisation (Toshima, Uematsuy and Hirahara 2003: 327–9; Hirahara, Sawada and Morikawa 2015: 159–66).

Simplified Head Models:

Abstracted head forms with minimal facial features, employed in comparative analyses of various electroacoustic recording systems (Palacino, Feichter and Rueff 2020: 2391–6; Vidal et al. 2021: 1–8; Grandjean et al. 2023).

Miniaturised Head Models:

Scaled-down versions of a human head, designed to test whether reduced dimensions affect the accuracy of spatial sound capture (Epain and Jin 2012: 91–102; Uchibori et al. 2015: 43–5).

Polyhedral Microphone Arrays:

Large-scale spherical arrays comprising multiple microphones symmetrically distributed around the sound source. These arrays are often sufficiently large to accommodate a performer within the array, capturing spatial sound fields from the inside out (Li and Duraiswami 2007: 702–14; Pollow, Behler and Masiero 2009; Hagaia et al. 2011: 2003–15).

Non-anthropomorphic Arrays:

Microphone systems based on various geometric shapes – such as spheres, ellipsoids and ovals – configured as open, closed, semi-open or hybrid arrays to suit different spatial audio applications (Kahana et al. 1999: 1503–16; Huang and Benesty 2004; Li and Duraiswami 2005: 1137–40; Balmages and Rafaely 2007: 727–32; Li and Duraiswami 2007: 702–14; Kubota et al. 2008: 468–76; Rafaely 2008a: 5–8; 2008b: 740–7; Zotkin, Duraiswami and Gumerov 2008: 277–80; Loufopoulos et al. 2014: 236–42; Loufopoulos, Heliades and Emmanouil 2015: 192–209; Dick and Vigeant 2016: 34–45; Fernandez Grande 2016: 1168–78; Dziwis et al. 2019: 883–5; Lee and Johnson 2021: 871–87).

3. Construction and description of authorial spherical microphone project

The spatial distribution of microphones across the surface of the prototype was designed to provide broad and balanced directional

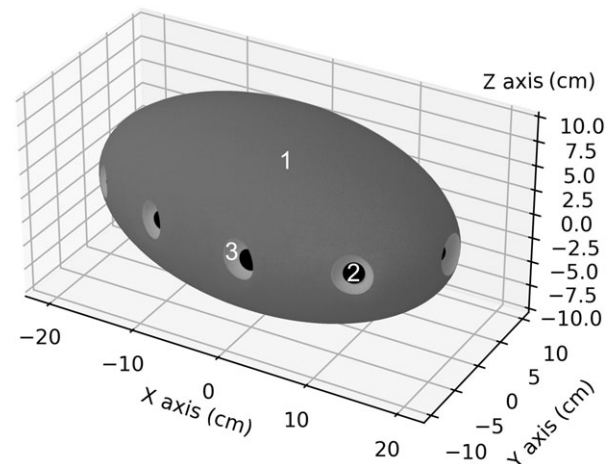


Figure 1. Detailed three-dimensional view of the ellipsoid microphone enclosure used in the spherical microphone prototype. The principal axes (X, Y, Z) are indicated in centimetres, illustrating the geometry that enables true spatial sampling for multichannel audio capture and acoustic research. The main body is labelled as 1; the position of an individual microphone capsule as 2; and the flexible gasket – allowing two-axis movement of the capsule while also protecting the device from dust and debris ingress – as 3.

coverage for spatial audio capture, with a particular emphasis on musical applications rather than strict theoretical optimisation for HOA. Unlike classical spherical arrays, where capsules are distributed over a perfect sphere, this design employs an ellipsoid casing (dimensions: $a = 20$ cm, $b = 10$ cm, $c = 10$ cm), mainly for ergonomic and practical reasons, as shown in Figure 1.

While the arrangement does not strictly conform to the mathematical ideal of equidistant spherical sampling required for high-order ambisonics, it was found to be effective in a wide range of artistic scenarios. The primary aim was to offer composers and sound artists a flexible and accessible tool for multichannel, immersive recording, rather than to replicate laboratory-grade measurement devices.

In the present prototype, ten condenser capsules are arranged uniformly along a single plane, forming an equatorial ring around the ellipsoid's surface. Each capsule is positioned at an approximately equal angular distance from its neighbours, thereby ensuring a consistent spatial distribution confined to the equator of the enclosure. While the theoretical minimum number of capsules required for second-order ambisonic recording is nine, the use of ten capsules was chosen for several practical reasons. First, a tenth capsule provides redundancy, reducing the risk of performance degradation in the event of a capsule failure during critical sessions. Second, the additional capsule allows for a more flexible spatial arrangement on the ellipsoid surface, mitigating spatial aliasing effects and improving the balance between ambisonic and binaural recording modes. Third, ten capsules offer greater symmetry and consistency in multichannel workflows, supporting more advanced rendering techniques and providing enhanced control for applications beyond the strict requirements of HOA theory (Zotkin, Duraiswami and Gumerov 2008: 277–80).

Outer casing:

The casing is fabricated from a high-quality plastic composite consisting of acrylonitrile-butadiene-styrene (ABS), polycarbonate and medical-grade silicone. ABS was selected for its high impact resistance and ease of moulding into the ellipsoidal shape. Polycarbonate contributes exceptional impact resistance and

transparency, providing both durability and aesthetic appeal – key factors in performance environments. Additionally, the casing is coated with a thin, transparent layer of medical-grade silicone to enhance grip and minimise the risk of accidental slippage. This hybrid material structure also supports the potential integration of sensors (e.g., accelerometers, infrared trackers) for orientation tracking in algorithmically choreographed sound installations (Waters 2011: 95–6; Bull and Back 2015).

It is important to emphasise that the prototyping stage represents the most costly phase of device development. Custom design, unique PCB fabrication, one-off enclosure production and the initial creation of control firmware all contribute significantly to overall expenses. These upfront costs can greatly exceed the material value of individual components, especially when manufacturing only a single or a small number of units. However, once the design is finalised and the project transitions into small-batch or serial production, the per-unit cost can be reduced substantially. This is achieved through economies of scale, streamlined assembly and the standardisation of manufacturing processes. Such transparency regarding the economic challenges of innovation is intended to foster collaborative development and facilitate broader adoption by independent artists, research laboratories and educational institutions.

Microphone capsules:

Each of the ten condenser capsules in the prototype is equipped with a motorised mechanism, allowing for remote adjustment of both angle and polar pattern in real time. The angular orientation of each capsule can be continuously varied within a range of -30° to $+30^\circ$ in both horizontal and vertical planes, providing composers and engineers with fine-grained control over the spatial focus of the array.

Each capsule is actuated by a miniature stepper motor, selected for its high precision, low noise and compact size. The motors are directly coupled to the capsule mounts via custom-designed gearing, enabling smooth bidirectional adjustment of angle in both axes. Motor control is achieved using microcontroller-driven pulse-width modulation, allowing for incremental, programmable changes in orientation. To suppress mechanical vibration and operational noise, each motor is isolated from the microphone chassis by a two-stage suspension system combining medical-grade silicone and high-density rubber bushings. Preliminary laboratory measurements indicate that the residual vibration transmitted to the capsules remains below the system's noise floor during standard operation, although further optimisation of the damping structure is ongoing. All motors are powered via a central control board, which communicates with the user interface (either the hardware control panel or software via USB) to synchronise and automate positioning across the array. This modular design enables maintenance and future upgrades of individual motor units without the need to disassemble the entire array.

Protective rubber element:

A rubberised seal – manufactured from a blend of medical-grade silicone and high-density rubber – encases the internal electronics. This element serves a dual purpose: it shields the circuitry from dust and moisture during microphone angle adjustments and provides passive acoustic insulation against external sounds. Its inherent flexibility ensures that the seal deforms under pressure and subsequently returns to its original shape without structural degradation.

Figure 2 schematically illustrates the potential angular adjustments of each of the ten microphones (front view) across two

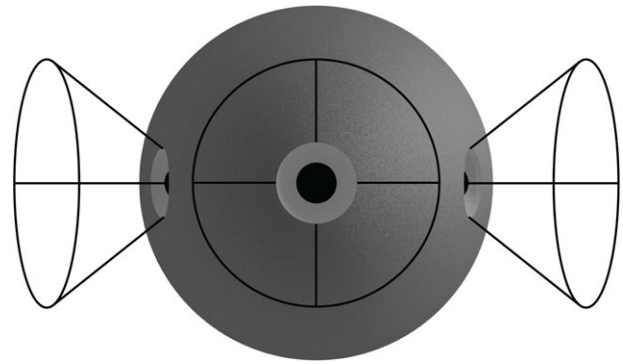


Figure 2. Frontal view of the ellipsoid microphone enclosure, showing the spatial orientation and schematic representation of the directional axes for the microphone capsules as their positions are adjusted. The wide angle of the illustrated axes simulates the range of motion available to the movable capsules; with fixed capsules, the axes would be considerably narrower. The diagram demonstrates the symmetrical capsule layout and its capacity for precise alignment in spatial audio recording.

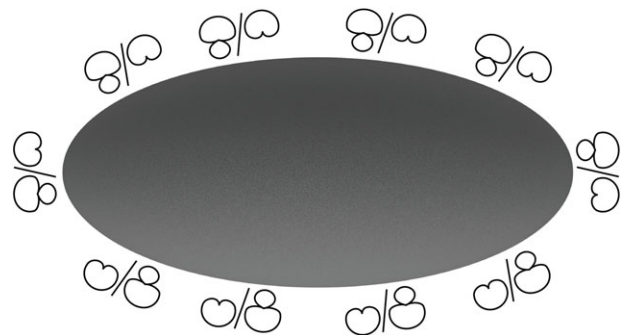


Figure 3. Top view of the ellipsoid microphone enclosure, illustrating the arrangement and two selectable polar patterns available for each microphone capsule. The cardioid and supercardioid icons indicate the individual directivity options that can be assigned independently by the user for each capsule.

planes. Each capsule can be adjusted within an angular range of -30° to $+30^\circ$ along both the horizontal and vertical axes. Empirical studies indicate that angular widths between 15° and 30° yield high localisation accuracy in human listeners when perceiving sound in multichannel audio systems (Zaunschirm, Frank and Zotter 2020). The microphone array employs a four-directional orientation mechanism, enabling each capsule to move independently in both the left–right and up–down directions. This directionality can be modified either manually, according to user-defined settings, or automatically via preset configurations provided by the device manufacturer.

From a top-view perspective, Figure 3 displays the precise spatial distribution of the ten condenser microphone capsules embedded within the casing. The diagram also indicates the available polar patterns – cardioid and supercardioid – which can be selected via an external control interface.

Polar patterns are selectable between cardioid and supercardioid modes, with switching implemented electronically via the control panel or software interface. At the present stage, the system supports discrete switching between these two patterns for each capsule, rather than continuous interpolation or morphing between polar responses. The underlying hardware design, however, leaves open the possibility for future firmware upgrades that might allow more fluid transitions or hybrid patterns.

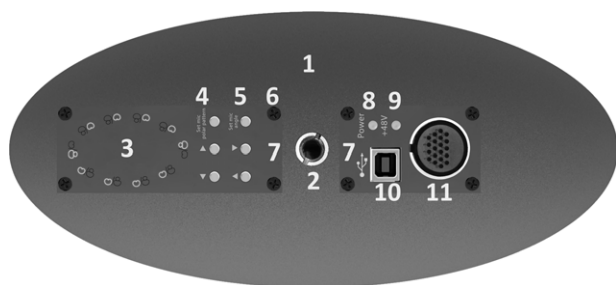


Figure 4. Bottom panel of the ellipsoid microphone enclosure with labelled components: (1) main enclosure; (2) tripod mount; (3) indicator LEDs for capsule polar pattern selection (active capsule highlighted); (4) buttons for polar pattern selection and individual capsule choice; (5) buttons for adjusting the orientation of each capsule; (6) mounting screw; (7) metal plate securing the buttons, sockets and indicator LEDs; (8) power/status indicator; (9) +48V phantom power indicator; (10) USB port; and (11) multipin audio connector. This configuration enables direct control of microphone parameters and convenient connections for power and audio signal.

A key innovation of this prototype is its ability to dynamically adjust both the angular orientation and the polar pattern of each microphone capsule in real time. By switching among cardioid, supercardioid or hybrid polar responses, composers can tailor the microphone's directivity to match the specific acoustical properties of a performance environment. This adaptability is particularly advantageous in site-specific installations and live performances, where spatial characteristics often vary across different locations or set-ups (Rafaely 2005: 135–43; Lee and Johnson 2021: 871–87).

Furthermore, the capacity for real-time reconfiguration of polar patterns opens up new creative possibilities in electroacoustic composition. The microphone's directivity profile can be algorithmically adjusted in response to musical or extramusical stimuli – such as data from motion sensors or generative patches – thereby allowing spatial sound properties to evolve dynamically alongside musical gestures (Landy 2007; Ramakrishnan 2009: 268–76). This feature enables compositional strategies in which the microphone array itself becomes an active, performative element within the work's spatial form.

Figure 4 illustrates the control panel designed for the spherical microphone array along with its associated connectors. The numbered labels correspond to the following components:

Microphone casing:

The outer casing is constructed from a durable plastic composite, ensuring both structural integrity and reliable acoustic performance.

Threaded adapter for mounting:

A universal threaded adapter (5/8-inch to 3/8-inch) enables secure mounting on a standard microphone stand.

Illuminated control panel:

An electronic display provides real-time visual feedback, indicating the selected polar pattern for each microphone capsule.

Horizontal adjustment buttons (left/right):

- When the polar pattern indicators display a green light, the user can cycle through the microphones (next/previous) to select the capsule for polar pattern adjustment.
- When the indicators display a red light, the left and right buttons adjust the microphone angle horizontally within a range of -30° to $+30^\circ$.
- The 'Set Mic Polar Pattern' button allows the user to assign the desired polar pattern (e.g., cardioid or supercardioid) to the selected microphone.

Vertical adjustment buttons (up/down):

These buttons facilitate vertical angle adjustments of each microphone within a range of -30° to $+30^\circ$.

Mounting screws:

Mounting screws secure both the control panel and the adjacent power supply unit, ensuring overall device stability during operation.

Metal panel:

The control panel and power supply unit are housed within a metal casing, which protects sensitive electronic components while providing thermal stability.

Blue LED indicator (power status):

A blue LED illuminates when the device is powered on, confirming its operational status.

Red LED indicator (phantom power):

- A solid red LED indicates that phantom power is active across all ten microphones.
- A flashing red LED signals that phantom power has not been activated for one or more microphone channels. If the polar pattern for a specific capsule does not illuminate or cannot be adjusted, it suggests that phantom power is unavailable for that microphone.

USB port:

The USB port enables connection to a computer for firmware updates, parameter adjustments and real-time microphone control. The control software is compatible with industry-standard digital audio workstations (DAWs) and interactive programming environments such as Max/MSP and SuperCollider, commonly used in electroacoustic composition and live performance.

Multipin audio connector:

This connector provides an analogue connection between the spherical microphone array and an external audio interface. The device is supplied with a dedicated cable featuring a converter that terminates in ten male XLR plugs, supporting both studio-based and mobile recording scenarios. This design addresses the needs of contemporary composers working in hybrid contexts that combine composition, performance and practice-based sound research (Bates 2009).

4. Directions of development

One of the most promising directions in contemporary audio technology is the integration of artificial intelligence (AI) and machine learning algorithms into audio processing workflows. These computational techniques enable advanced sound analysis and synthesis, facilitating the creation of increasingly complex and realistic auditory environments (Geier et al. 2010: 219–27). By leveraging pattern recognition and adaptive filtering, AI-driven algorithms can dynamically optimise microphone configurations, thereby enhancing spatial precision and realism in immersive audio applications.

It is important to clarify that the microphone prototype itself does not contain built-in AI or autonomous signal processing capabilities. Instead, AI-driven optimisation and generative control are implemented at the software level, using external platforms such as Max/MSP, SuperCollider or custom DAW plug-ins. The microphone transmits control data and receives configuration commands via its USB interface, allowing users to integrate machine learning algorithms and real-time sensor data into their creative workflow. This separation of hardware and

software ensures maximum flexibility, enabling users to adopt the latest computational tools without requiring hardware modifications. As a result, the device functions as a responsive, reconfigurable front-end within a broader, user-defined ecosystem for spatial audio creation and research.

In parallel, adaptive microphone systems – such as those incorporating motorised capsules capable of adjusting their orientation in response to environmental changes – represent a significant innovation in recording technology. By automatically modifying microphone angles relative to the sound source or acoustic environment, these systems improve recording accuracy, particularly during initial calibration procedures. This approach has the potential to significantly heighten the immersive experience in VR applications, video games and interactive installations.

The growing relevance of immersive audio technologies is also reflected in the expanding creative and commercial landscapes of VR, AR and MR. These formats increasingly rely on precise spatial audio reproduction to foster perceptual realism, thereby opening new avenues for artistic and technological development. Advanced spherical microphone arrays capable of real-time reconfiguration support these efforts by providing sound designers and electroacoustic composers with versatile tools for crafting intricate spatial narratives.

Beyond artistic applications, these technologies offer considerable potential for scientific inquiry. Spherical microphone arrays equipped with adjustable capsules can be employed to study spatial acoustic fields, providing valuable data for research in architectural acoustics, psychoacoustics and environmental sound monitoring. The ability to capture spatially distributed acoustic information with high resolution makes these devices suitable for tasks ranging from concert hall optimisation to immersive telepresence systems.

Notably, most commercially available spherical microphone arrays lack motorised, position-adjustable capsules – a design limitation that the present prototype aims to overcome. The ability to modify microphone angles and select capsule-specific polar patterns in real time significantly enhances the device's adaptability across diverse acoustic contexts. This versatility renders the system equally suitable for recording studios, large-scale live performances and film sets, where tailoring microphone configurations to specific spatial conditions is essential for achieving optimal results.

Much like the introduction of stereo sound and, later, digital audio technologies reshaped musical practices in the twentieth century, the advent of HOA and advanced binaural recording techniques is now influencing twenty-first-century electroacoustic composition and performance (Landy 2007; Wishart 1996). The proposed spherical microphone, with its capacity for real-time angular adjustments and dynamically adjustable polar patterns, aims to foster a more nuanced approach to spatial recording. By offering enhanced control over the spatial attributes of captured sound fields, the device supports a wide range of artistic outcomes – from headphone-based acousmatic compositions to expansive, multispeaker installations in site-specific contexts.

From a compositional perspective, the availability of advanced spatial capture tools may catalyse the development of new artistic paradigms. Electroacoustic composers are increasingly engaging with multichannel sound diffusion techniques, head-tracked headphone listening and hybrid MR environments, all of which depend on accurate spatial imaging (Emmerson 2007; Grimshaw 2014; Jerald 2015). The broader literature documents numerous examples of creative practitioners exploring immersive sound as a

means to reconfigure traditional listening practices – from bespoke acousmatic concerts employing custom multispeaker arrays (Bates 2009) to networked performances that connect multiple physical and virtual venues in real time (Bull and Back 2015; Jerald 2015).

By combining technological innovation with creative flexibility, this spherical microphone prototype contributes to ongoing research into immersive audio systems while simultaneously offering new aesthetic and compositional possibilities for electroacoustic music and sound art.

5. Electroacoustic composition and sound design

Immersive sound has become an increasingly integral component of electroacoustic composition, where spatial architecture serves not merely as a presentation layer but as a core structural element woven into a work's formal and aesthetic design (Baalman 2010: 209–18; Wishart 1996; Harrison 1998: 117–27; Landy 2007). Techniques such as ambisonic layering, multichannel diffusion and hybrid headphone–loudspeaker configurations empower composers to craft intricate sonic trajectories and evolving spatial perspectives. By offering a single device that captures both second-order ambisonic information and binaural cues, the proposed spherical microphone prototype alleviates the logistical challenges associated with deploying multiple microphone arrays simultaneously (Kendall 2010: 228–38).

The potential applications of the prototype span several key domains in electroacoustic composition and sound design:

Acousmatic Environments:

The ability to capture environmental audio from varying angles and with diverse polar patterns enables the creation of layered recordings for multichannel diffusion. By selectively adjusting microphone positions and polar responses, composers can accentuate or attenuate specific spectral components within a soundscape. This flexibility facilitates the dynamic sculpting of acousmatic textures during real-time performances or studio-based compositions. For instance, a soundscape recorded in an urban environment might emphasise high-frequency pedestrian chatter at one moment, while foregrounding low-frequency vehicular rumble at another, thus guiding the listener's attention through deliberate spatial manipulation.

Soundscape Composition:

In approaches rooted in ecological and place-based practices, soundscape composition often relies on field recordings as foundational material (Truax 1992: 37–40; McCartney 2002: 45–9; Moore 2016). The prototype's motorised microphone capsules, with their capacity for dynamic orientation, enable the selective amplification of environmental elements – such as rain patterns, avian calls or distant machinery – at predetermined angles and intervals. This capability supports the creation of immersive soundscapes where spatial characteristics contribute to narrative form, thereby fostering deeper perceptual engagement with the ecological and cultural context of the recorded environment.

Augmented Instrument Recording:

The spherical microphone prototype also opens new avenues for capturing ensemble and solo performances, particularly when dynamic directivity is crucial. By adjusting the angle and polar pattern of individual microphone capsules during a performance, composers and sound engineers can highlight microsonic details, extended instrumental techniques and subtle performer movements with high spatial accuracy. This functionality is especially beneficial for chamber music practices that integrate live electronics (Normandeau 2009: 277–85; Bates 2009; Ramakrishnan 2009;

268–76). For example, a contemporary string quartet might employ the microphone to track nuanced timbral shifts and variations in bow pressure during passages played *sul ponticello*, thereby enhancing the immersive experience for both live and remote audiences.

6. Immersive installations, sound art and VR

For artists and curators engaged in large-scale sound installations or VR-based projects, the spherical microphone's real-time reorientation capabilities open up novel interactive possibilities (Emmerson 2007; Jerald 2015). The device can be programmed to track audience movement via integrated motion sensors, allowing it to focus on proximate sound sources or adapt dynamically to changes in the acoustic environment. This adaptive functionality introduces a degree of reactivity that is highly valued in interactive art, effectively transforming the technological infrastructure into an active participant in the emerging sonic narrative (McCartney 2002: 45–9).

Headphone-based installations similarly benefit from the microphone's ability to capture both ambisonic and binaural streams simultaneously. By ensuring consistency between on-site and remote listening conditions, artists can design parallel immersive experiences across different media platforms. This dual-stream functionality aligns with established approaches in immersive sound art, where both co-present and geographically dispersed audiences experience synchronous auditory events (Blessner and Salter 2006; Bates 2009).

7. Live performance and interactive music

Live electronic music, improvisatory practices and multiperformer scenarios increasingly incorporate sophisticated spatial recording strategies into the performance apparatus (Roads 2001; Collins, Schedel and Wilson 2013). The motorised capsules of the spherical microphone enable sound engineers and composers to shift 'listening perspectives' in real time, reacting to performer cues or generative algorithmic instructions. This capability enhances the interactive dimension of performances by extending creative agency to both performers and audiences (Bates 2009; Robusté 2018: 296–307; Waters 2011: 95–6).

Moreover, the prototype's dual recording capability – simultaneously capturing ambisonic and binaural streams – facilitates hybrid concert formats. In these contexts, in-person audiences experience multichannel loudspeaker diffusion, while remote listeners, equipped with headphones, enjoy an immersive binaural representation of the performance. This dual-mode set-up is particularly advantageous for networked music performances, where participants located in different physical spaces share a common auditory environment (Grimshaw 2014; Bull and Back 2015; Jerald 2015).

Additionally, site-specific installations and interactive musical works can further benefit from the microphone's ability to reposition capsules in response to performer trajectories, sensor triggers or audience interactions. By integrating the microphone with software platforms such as Max/MSP, SuperCollider and other interactive environments, artists can design performances where the spatial characteristics of the captured sound field evolve organically in response to external stimuli.

8. Artistic workflow and use scenarios

The spherical microphone prototype was designed with the creative workflow of composers and sound artists in mind, providing tools that directly inspire new approaches to spatial composition, performance and sound design. Unlike conventional arrays, this system allows the user to reconfigure the spatial focus and directivity of each capsule in real time, either manually during rehearsal and performance or algorithmically in response to live data streams (such as motion tracking or musical parameters).

In a typical workflow, an artist might deploy the microphone in a complex acoustic space – for example, a reverberant gallery or abandoned industrial site – where the acoustic signature of the environment becomes a compositional parameter. The capsules can be dynamically oriented to follow soloists or ensembles as they move through the space, enabling the recording of highly immersive, evolving spatial soundscapes. This flexibility eliminates the need for multiple static microphone arrays and complex post-production, as spatial gestures can be 'performed' directly during the recording.

In live electroacoustic performance, the ability to switch directivity or reorient capsules in response to musical cues allows for truly interactive spatialisation, where the performer, sound engineer or algorithm can shape the perceived sound field in real time. This opens up new modes of audience engagement – for instance, by focusing the array towards areas of greatest musical activity or by intentionally reshaping the ambient soundscape to match the dramaturgy of a performance or installation.

For composers working in studio contexts, the device simplifies the workflow for capturing multiperspective source material for acousmatic, VR or binaural projects. By adjusting the physical configuration of the capsules *in situ*, the artist can tailor the recorded material for later spatialisation and diffusion without relying on digital post-processing alone.

Finally, the system's dual-output architecture enables the simultaneous capture of high-order ambisonic signals for loudspeaker diffusion and binaural feeds for remote or headphone-based audiences. This supports new hybrid concert formats and distributed sound art projects, where both local and online listeners can share in a coherent immersive experience.

In summary, the microphone is not only a recording tool but also a performative instrument in its own right, capable of transforming the creative workflow and expanding the expressive vocabulary available to today's composers, performers and sound designers.

9. Discussion

The development of spherical microphone arrays – particularly those integrating motorised capsules and flexible directivity – raises a complex set of both technical and artistic issues. From a design perspective, the use of a rigid ellipsoid casing offers enhanced durability and a stable acoustic reference, facilitating consistent and repeatable spatial sound capture (Meyer and Elko 2002: 1781–4; Rafaely 2005: 135–43). Yet this approach also introduces potential challenges, such as mechanical complexity, the risk of unwanted scattering effects and the need to balance structural robustness with the minimisation of mechanical noise (Dziwis et al. 2019: 883–5).

Artistically, the prototype's real-time configurability and multidimensional spatial coverage empower composers and

performers to approach space as a dynamic parameter within the creative process (Bates 2009; Landy 2007). By allowing the physical orientation and directivity of each capsule to be adapted during a performance or recording, the device fosters new methods of interactive spatialisation, supports responsive and evolving soundscapes and encourages cross-disciplinary experimentation – whether in acousmatic music, live electronic performance or site-specific installations (Smalley 1997: 107–26; Baalman 2010: 209–18; Emmerson 2007).

Despite these advances, certain limitations remain. The physical arrangement of capsules on an ellipsoid, while effective in most artistic contexts, does not strictly conform to the ideal mathematical models of HOA arrays (Abhayapala and Ward 2002: 1949–52). This may constrain the spatial resolution achievable in highly analytical or scientific applications. Additionally, the ongoing refinement of the motorisation system – especially in terms of vibration and noise isolation – will be crucial to ensuring the device's reliability in the most demanding acoustic environments.

Ultimately, the prototype represents a balancing act between theoretical optimisation and practical creativity. Its design is shaped as much by the needs of artists as by the requirements of measurement science, with a deliberate emphasis on usability, flexibility and the encouragement of new artistic workflows (Bates 2009; Wishart 1996). The collaborative, iterative approach adopted in the project points towards a future in which innovative recording tools are developed not in isolation but in close dialogue with the communities of practice they are intended to serve.

10. Conclusion

The spherical microphone prototype presented in this study addresses a significant gap in the market for versatile audio recording devices. By integrating ambisonic, binaural and immersive recording capabilities into a single unit, the device offers a comprehensive solution tailored to the diverse needs of contemporary audio professionals. The core innovation lies in its ellipsoid-shaped casing, which houses ten motorised condenser microphones whose angles can be adjusted relative to the sound source or recording environment. Additionally, selectable polar patterns enhance the precision and realism of spatial audio capture, while the use of second-order ambisonics ensures high-resolution sound field recording – supporting both creative and scientific endeavours.

A review of current literature and available technologies reveals that while spherical and dummy-head microphones are commonly used, no existing solution matches the versatility of this prototype. Traditional designs tend to be limited to specific applications, whereas this device integrates features that meet a wide range of spatial recording requirements. Its potential applications extend well beyond the entertainment industry – including video games, VR and immersive media – encompassing fields such as scientific research, acoustic engineering and phonography. The prototype's ability to capture studio performances, live stage events and on-location sound for film production demonstrates its broad utility and adaptability.

Beyond its artistic applications, the spherical microphone also provides a valuable tool for research on auditory scene analysis, particularly in studies of perceptual streaming. Because the device enables precise control of spatial cues and dynamic reconfiguration of capsule orientation, it can be used to capture sound environments in which listeners perceive the emergence, separation and fusion of auditory streams. This creates new possibilities for documenting spatially complex textures relevant to classical psychoacoustic paradigms, including those developed by Albert S.

Bregman in his foundational work on auditory organisation and stream segregation (Bregman, 1990).

Furthermore, the prototype's real-time adjustability introduces unprecedented operational flexibility. Users can dynamically configure microphone positions to optimise sound capture in varying acoustic environments and performance contexts. This adaptability is particularly beneficial for recording complex soundscapes with moving sources or in unpredictable settings. The incorporation of motorised capsules into the spherical array simplifies technical set-up procedures, reducing the need for multiple devices or cumbersome rigging and thereby enhancing workflow efficiency for sound engineers.

The potential for further innovation remains considerable. Future iterations could integrate machine learning algorithms for automated microphone positioning based on real-time acoustic analysis, as well as software enhancements to support interactive sound installations. By bridging technological innovation with artistic and scientific applications, this spherical microphone prototype contributes to the evolving landscape of immersive audio. It represents a significant step towards more nuanced and precise sound reproduction across professional, academic and creative domains, encouraging new explorations in the art and science of spatial sound.

11. Future research and adjustments

The ongoing development of the spherical microphone prototype is guided by a commitment to both technical excellence and artistic relevance. Key priorities for future work include comprehensive acoustic testing – in controlled environments and real-world scenarios – to ensure the device's reliability, adaptability and sonic transparency across diverse spatial contexts. Particular emphasis will be placed on verifying the long-term stability of the motorised mechanisms and further reducing any residual mechanical noise.

Beyond technical refinement, a central strand of research will focus on artistic collaboration and feedback: inviting composers, sound artists and performers to test the prototype in creative projects, public showcases and educational workshops. Insights gathered from these applications will directly inform further hardware and software improvements, as well as inspire new functionalities tailored to the evolving needs of spatial music practitioners.

Looking ahead, future iterations may explore the integration of advanced features such as algorithmic or AI-driven capsule control, enhanced compatibility with emerging audio platforms and expanded interactive capabilities for immersive installations and live performance. Ultimately, the project aims to foster an ongoing dialogue between technological innovation and creative exploration, positioning the microphone as both a robust recording instrument and a catalyst for artistic discovery.

References

- Abhayapala, T. D. and Ward, D. B. 2002. Theory and Design of High Order Sound Field Microphones Using Spherical Microphone Array. *IEEE International Conference on Acoustics, Speech, and Signal Processing*. Orlando, FL, USA, 1949–52. <https://doi.org/10.1109/ICASSP.2002.5745011>
- Baalman, M. A. J. 2010. Spatial Composition Techniques and Sound Spatialisation Technologies. *Organised Sound* 15(3): 209–18. <https://doi.org/10.1017/S1355771810000245>
- Balmages, I. and Rafaely, B. 2007. Open-Sphere Designs for Spherical Microphone Arrays. *IEEE Transactions on Audio, Speech, and Language Processing* 15(2): 727–32. <https://doi.org/10.1109/TASL.2006.881671>

- Bates, E. 2009. The Composition and Performance of Spatial Music. Doctoral thesis, University of Dublin. <http://www.endabates.net/Enda%20Bates%20-%20The%20Composition%20and%20Performance%20of%20Spatial%20Music.pdf> (accessed 18 March 2025)
- Blessner, B. and Salter, L. 2006. Spaces Speak, Are You Listening? Experiencing Aural Architecture. Cambridge, MA: MIT Press. <https://doi.org/10.7551/mitpress/6384.001.0001>
- Bregman, A. S. 1990. *Auditory Scene Analysis: The Perceptual Organization of Sound*. Cambridge, MA: MIT Press. <https://doi.org/10.7551/mitpress/1486.001.0001>
- Bull, M. and Back, L. (eds.) 2015. *The Auditory Culture Reader* (2nd ed.). London: Routledge. <https://doi.org/10.4324/9781003086895>
- Collins, N., Schedel, M. and Wilson, S. 2013. *Electronic Music*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9780511820540>
- Dick, D. A. and Vigeant, M. C. 2016. A Comparison of Measured Room Acoustics Metrics Using a Spherical Microphone Array and Conventional Methods. *Applied Acoustics* **107**: 34–45. <https://doi.org/10.1016/j.apacoust.2016.01.008>
- Dziwis, D. T., Lübeck, T., Arendt, J. M. and Pörschmann, Ch. 2019. Development of a 7th Order Spherical Microphone Array for Spatial Audio Recording. *Conference: Fortschritte der Akustik – DAGA 2019 Rostock*. DEGA e.V., D-Berlin, 883–5.
- Emmerson, S. 2007. *Living Electronic Music*. Ashgate/London: Routledge. <https://doi.org/10.4324/9781351217866>
- Epain, N. and Jin, C. T. 2012. Independent Component Analysis Using Spherical Microphone Arrays. *Acta Acustica United With Acustica* **98**: 91–102. <https://doi.org/10.3813/AAA.918495>
- Fernandez Grande, E. 2016. Sound Field Reconstruction Using a Spherical Microphone Array. *The Journal of the Acoustical Society of America* **139**(3): 1168–78. <https://doi.org/10.1121/1.4943545>
- Findlay-Walsh, I. 2017. Sonic Autoethnographies: Personal Listening as Compositional Context. *Organised Sound* **23**(1): 121–30. <https://doi.org/10.1017/S1355771817000371>
- Gardner, B. and Martin, K. 1994. HRTF Measurements of a KEMAR Dummy-Head Microphone. *MIT Media Lab Perceptual Computing – Technical Report #280*.
- Geier, J., Ahrens, J. and Spors, S. 2010. Object-Based Audio Reproduction and the Audio Scene Description Format. *Organised Sound* **15**(3): 219–27. <https://doi.org/10.1017/S1355771810000324>
- Grandjean, P., Robin, O., Berry, A. and Gauthier, Ph.-A. 2023. Evaluation of the Respective Effects of Microphone Type and Dummy Head Type on Measured Head-Related Transfer Functions. *INTER-NOISE and NOISE-CON Congress and Conference Proceedings, InterNoise23*. Chiba, Japan, https://doi.org/10.3397/IN_2023_0300
- Grimshaw, M. (ed.) 2014. *The Oxford Handbook of Virtuality*. Oxford: Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780199826162.001.0001>
- Hagaia, I. B., Pollow, M., Vorländer, M. and Rafaely, B. 2011. Acoustic Centering of Sources Measured by Surrounding Spherical Microphone Arrays. *The Journal of the Acoustical Society of America* **130**(4): 2003–15. <https://doi.org/10.1121/1.3624825>
- Hardera, S., Paulsen, R. R., Larsen, M., Laugesen, S., Mihocic, M. and Majdak, P. 2006. A Framework for Geometry Acquisition, 3-D Printing, Simulation, and Measurement of Head-Related Transfer Functions with a Focus on Hearing-Assistive Devices. *Computer-Aided Design* **75–76**: 39–46. <https://doi.org/10.1016/j.cad.2016.02.006>
- Harrison, J. 1998. Sound, Space, Sculpture: Some Thoughts on the 'What', 'How' and 'Why' of Sound Diffusion. *Organised Sound* **3**(2): 117–27. <https://doi.org/10.1017/S1355771898002040>
- Hirahara, T., Sawada, Y. and Morikawa, D. 2015. Sound Localization of Dynamic Binaural Signals Provided Using a Pinna-Less Dummy Head or a Stereo Microphone. *Interdisciplinary Information Sciences* **21**(2): 159–66. <https://doi.org/10.4036/iis.2015.A.07>
- Huang, Y. and Benesty, J. (eds.) 2004. *Audio Signal Processing for Next Generation Multimedia Communication Systems*. New York, Boston, Dordrecht, London, Moscow: Kluwer Academic Publishers.
- Jerald, J. 2015. *The VR Book: Human-Centered Design for Virtual Reality*. New York: ACM & Morgan & Claypool. <https://doi.org/10.1145/2792790>
- Kahana, Y., Nelson, Ph. A., Kirkeby, O. and Hamada, H. 1999. A Multiple Microphone Recording Technique for the Generation of Virtual Acoustic Images. *The Journal of the Acoustical Society of America* **105**(3): 1503–16. <https://doi.org/10.1121/1.426690>
- Kendall, G. S. 2010. Spatial Perception and Cognition in Multichannel Audio for Electroacoustic Music. *Organised Sound* **15**(3): 228–38. <https://doi.org/10.1017/S1355771810000336>
- Kubota, Y., Yoshida, M., Komatani, K., Ogata, T. and Okuno, H. G. 2008. Design and Implementation of 3D Auditory Scene Visualizer towards Auditory Awareness With Face Tracking. *Tenth IEEE International Symposium on Multimedia*, Berkeley, CA, USA, 468–76. <https://doi.org/10.1109/ISM.2008.107>
- Landy, L. 2007. *Understanding the Art of Sound Organization*. Cambridge, MA: MIT Press. <https://doi.org/10.7551/mitpress/7472.001.0001>
- Lee, H. and Johnson, D. 2021. 3D Microphone Array Comparison: Objective Measurements. *Journal of the Audio Engineering Society* **69**(11): 871–87. <https://doi.org/10.17743/jaes.2021.0038>
- Li, Z. and Duraiswami, R. 2005. A Robust and Self-Reconfigurable Design of Spherical Microphone Array for Multi-Resolution Beamforming. *Proceedings. (ICASSP '05). IEEE International Conference on Acoustics, Speech, and Signal Processing*. Philadelphia, PA, USA, 4: 1137–40. <https://doi.org/10.1109/ICASSP.2005.1416214>
- Li, Z. and Duraiswami, R. 2007. Flexible and Optimal Design of Spherical Microphone Arrays for Beamforming. *IEEE Transactions on Audio, Speech, and Language Processing* **15**(2): 702–14. <https://doi.org/10.1109/TASL.2006.876764>
- Loufopoulos, A., Heliades, G. and Emmanouil, M. 2015. An Innovative Audiovisual Immersion Space and Its Prospects for Cultural Heritage Advancement. *International Journal of Computational Intelligence Studies* **4**(2): 192–209. <https://doi.org/10.1504/IJCISTUDIES.2015.071186>
- Loufopoulos, A., Heliades, G., Emmanouil, M. and Matagkos, F. 2014. Realism, Art, Technology and Audiovisual Immersion into the Environment of the Ionian Islands. In A. Georgaki and G. Kouroupetroglou (eds.) *Proceedings ICMC|SMC|2014*, Athens, Greece, 14–20 Sep., 236–42.
- McCartney, A. 2002. Alien Intimacies: Hearing Science Fiction Narratives in Hildegard Westerkamp's Cricket Voice (or 'I Don't Like the Country, the Crickets Make Me Nervous'). *Organised Sound* **7**(1): 45–9. <https://doi.org/10.1017/S1355771802001073>
- Meyer, J. and Elko, G. 2002. A Highly Scalable Spherical Microphone Array Based on an Orthonormal Decomposition of the Soundfield. *IEEE International Conference on Acoustics, Speech, and Signal Processing*, Orlando, FL, USA, 1781–4. <https://doi.org/10.1109/ICASSP.2002.5744968>
- Mobashsher, A. T. and Abbosh, A. M. 2014. Three-Dimensional Human Head Phantom With Realistic Electrical Properties and Anatomy. *IEEE Antennas and Wireless Propagation Letters* **13**: 1401–4. <https://doi.org/10.1109/LAWP.2014.2340409>
- Moore, A. 2016. *Sonic Art An Introduction to Electroacoustic Music Composition*. New York: Routledge. <https://doi.org/10.4324/9781315684024>
- Normandeau, R. 2009. Timbre Spatialisation: The Medium is the Space. *Organised Sound* **14**(3): 277–85. <https://doi.org/10.1017/S1355771809990094>
- North, K. 2024. The Haptic in Soundscape Composition. *Organised Sound* **29**(3): 292–302. <https://doi.org/10.1017/S135577182400027X>
- O'Connor, D. and Kennedy, J. 2021. An Evaluation of 3D Printing for the Manufacture of a Binaural Recording Device. *Applied Acoustics* **171**: 107610. <https://doi.org/10.1016/j.apacoust.2020.107610>
- Obadia, J. 2021. The Application of Binaural System on Virtual Reality Technology Through the Animation of Wayang Kulit Yogyakarta Performance. *IMOVICCON Conference Proceeding* **2**(1): 87–94. <https://doi.org/10.37312/imoviccon.v2i1.97>
- Palacino, J., Feichter, L. and Rueff, P. 2020. Acoustic Impact of Binaural In-Ear Microphone Positioning. *Forum Acusticum*, 2391–6. <https://doi.org/10.48465/fa.2020.0322>
- Pollow, M., Behler, G. and Masiero, B. 2009. Measuring Directivities of Natural Sound Sources with a Spherical Microphone Array. *Ambisonics Symposium*, Graz, Austria, 25–27 June.
- Rafaely, B. 2005. Analysis and Design of Spherical Microphone Arrays. *IEEE Transactions on Speech and Audio Processing* **13**(1): 135–43. <https://doi.org/10.1109/TSA.2004.839244>
- Rafaely, B. 2008a. Spatial Sampling and Beamforming for Spherical Microphone Arrays. *Hands-Free Speech Communication and Microphone Arrays*, Trento, Italy, 5–8. <https://doi.org/10.1109/HSCMA.2008.4538673>

- Rafaely, B. 2008b. The Spherical-Shell Microphone Array. *IEEE Transactions on Audio, Speech, and Language Processing* **16**(4): 740–7. <https://doi.org/10.1109/TASL.2008.920059>
- Ramakrishnan, C. 2009. Zirkonium: Non-Invasive Software for Sound Spatialisation. *Organised Sound* **14**(3): 268–76. <https://doi.org/10.1017/S1355771809990082>
- Rebelo, P. 2003. Performing Space. *Organised Sound* **8**(2): 181–6. <https://doi.org/10.1017/S1355771803000086>
- Roads, C. 2001. *Microsound*. Cambridge, MA: MIT Press. <https://doi.org/10.7551/mitpress/4601.001.0001>
- Robusté, J. R. 2018. Filling Sound with Space: An Elemental Approach to Sound Spatialisation. *Organised Sound* **15**(3): 296–307. <https://doi.org/10.1017/S1355771818000213>
- Smalley, D. 1997. Spectromorphology: Explaining Sound-Shapes. *Organised Sound* **2**(2): 107–26. <https://doi.org/10.1017/S1355771897009059>
- Smalley, D. 2007. Space-Form and the Acousmatic Image. *Organised Sound* **12**(1): 35–58. <https://doi.org/10.1017/S1355771807001665>
- Ting, T. M., Ahmad, N. S., Goh, P. and Mohamad-Saleh, J. 2021. Binaural Modelling and Spatial Auditory Cue Analysis of 3D-Printed Ears. *Sensors* **21**: 227. <https://doi.org/10.3390/s21010227>
- Toshima, I., Uematsuy, H. and Hirahara, T. 2003. A Steerable Dummy Head that Tracks Three-Dimensional Head Movement: TeleHead. *Acoustical Science and Technology* **24**(5): 327–9. <https://doi.org/10.1250/ast.24.327>
- Truax, B. 1992. Composing with Time-Shifted Environmental Sound. *Leonardo Music Journal* **2**(1): 37–40. <https://doi.org/10.2307/1513207>
- Uchibori, S., Sarumaru, Y., Ashihara, K., Ohta, T. and Hiryu, S. 2015. Experimental Evaluation of Binaural Recording System Using a Miniature Dummy Head. *Acoustical Science and Technology* **36**(1): 43–5. <https://doi.org/10.1250/ast.36.42>
- Vidal, A., Herzog, Ph., Lambourg, Ch. and Chatron, J. 2021. HRTF Measurements of Five Dummy Heads at Two Distances. *Immersive and 3D Audio: From Architecture to Automotive (I3DA)*, Bologna, Italy, 1–8. <https://doi.org/10.1109/I3DA48870.2021.9610914>
- Waters, S. 2011. Editorial: Performance Ecosystems. *Organised Sound* **16**(2): 95–6. <https://doi.org/10.1017/S1355771811000021>
- Wishart, T. 1996. *On Sonic Art. (New and revised edition edited by S. Emmerson)*. Harwood Academic Publishers/New York Routledge. <https://doi.org/10.4324/9781315077895>
- Yan, S., Sun, H., Svensson, U. P. and Hovem, J. M. 2011. Optimal Modal Beamforming for Spherical Microphone Arrays. *IEEE Transactions on Audio, Speech, and Language Processing* **19**(2): 361–71. <https://doi.org/10.1109/TASL.2010.2047815>
- Young, J. 2007. Reflections on Sound Image Design in Electroacoustic Music. *Organised Sound* **12**(1): 25–33. <https://doi.org/10.1017/S1355771807001689>
- Zaunschirm, M., Frank, M. and Zotter, F. 2020. Binaural Rendering with Measured Room Responses: First-Order Ambisonic Microphone versus Dummy Head. *Applied Sciences* **10**: 1631. <https://doi.org/10.3390/app10051631>
- Zotkin, D. N., Duraiswami, R. and Gumerov, N. A. 2008. Sound Field Decomposition Using Spherical Microphone Arrays. *IEEE International Conference on Acoustics, Speech and Signal Processing*, Las Vegas, NV, USA, 277–80. <https://doi.org/10.1109/ICASSP.2008.4517600>